EnterpriseOS

Business Requirements Document (BRD) + Software Requirements Specification (SRS)

Phase 4: Enterprise Workflows, Collaboration and Reporting

Document control	Value
Version	1.0 - Client review draft
Status	Proposed; client confirmation required
Project	EnterpriseOS - multi-tenant SaaS ERP
Prepared for	Client / Business Sponsor
Stack	React + TypeScript; Java Spring Boot; MySQL 8.4 LTS
Prerequisite	Approved and delivered Phase 3 baseline
Date	11 November 2026
Approval	Pending client sign-off

Purpose: Determine client decisions, specify the proposed phase scope and provide verifiable approval and acceptance gates. Any default described here is a proposal, not a confirmed client requirement.

1. Executive Summary & Business Outcomes

Add configurable process orchestration, messaging, document controls, rich reports and integration surfaces while preserving strict authorizations and auditable business events.

Phase 4 consumes the shared identity, tenancy and role model defined in Phase 1. Functional changes are implemented as new bounded modules in the same Spring Boot modular monolith. Existing security controls and audit records remain mandatory.

2. Prerequisites, Dependencies & Stakeholders

Dependency	Entry requirement
Prior phase	Phase 3 released, accepted and regression-tested.
Tenant & identity	Active tenant context, membership validation, session security and permission model available.
Reference data	Departments, staff, currencies, policies and other dependent master data approved as applicable.
Stakeholders	Client sponsor, product owner, department SMEs, information security contact, UAT owner nominated.
Delivery assets	Source repository, staging environment, MySQL migration process, CI tests, email service available.

Responsible stakeholders: Workflow Designer, Process Owner, Approver, Department Head, Reporting Analyst, Integration Administrator, Compliance Auditor.

3. Client Discovery - Required Inputs Before Scope Freeze

1. Which specific business workflows must be configurable by admins?
   Client answer / reference: _____

2. What decisions determine routing: amount, branch, category, reporting hierarchy or SLA?
   Client answer / reference: _____

3. Are approval steps sequential, parallel, delegated or consensus-based?
   Client answer / reference: _____

4. Which notifications require email vs in-app vs live UI delivery?
   Client answer / reference: _____

5. What are file type, size, access, retention and virus scanning requirements?
   Client answer / reference: _____

6. Which exact dashboards, report filters, export formats and refresh cadence are needed?
   Client answer / reference: _____

7. Which external systems use webhooks or API keys, and what data may be shared?
   Client answer / reference: _____

8. What audit retention, data deletion and compliance reporting are required?
   Client answer / reference: _____

4. Functional Requirements - Traceability Baseline

4.1 Configurable Workflow & Approval Engine

- P4-FR-0101 | Design versioned approval templates for supported business objects
- P4-FR-0102 | Support sequential/parallel steps, amount conditions and approver groups
- P4-FR-0103 | Track state transitions, deadlines, delegations and escalations
- P4-FR-0104 | Prevent duplicate decisions and self-approval using configurable policy
- P4-FR-0105 | Log each action with actor, timestamp, reason and version

4.2 Notifications & Internal Communication

- P4-FR-0201 | Create in-app notification center and per-event preferences
- P4-FR-0202 | Deliver transactional email notifications via verified provider
- P4-FR-0203 | Stream authorized updates using WebSocket or server-sent events
- P4-FR-0204 | Implement read/unread state and user notification settings
- P4-FR-0205 | Queue retries, suppress duplicates and log delivery outcomes

4.3 Documents & Controlled Files

- P4-FR-0301 | Upload, validate, classify and version company documents
- P4-FR-0302 | Apply tenant-scoped and object-scoped access policies
- P4-FR-0303 | Store blob data outside relational database and retain metadata in MySQL
- P4-FR-0304 | Support expiring signed download links and retention rules
- P4-FR-0305 | Maintain access/change history and malware-scanning process

4.4 Advanced Reporting & Integrations

- P4-FR-0401 | Deliver role-aware HR, CRM, procurement and finance dashboards
- P4-FR-0402 | Provide saved filters, scheduled exports and report snapshots
- P4-FR-0403 | Publish documented versioned REST APIs and event webhooks
- P4-FR-0404 | Implement token rotation, rate limiting and callback retry strategy
- P4-FR-0405 | Show operational indicators for failed jobs, workflow SLA and delivery status

P0 = mandatory only after signed baseline. P1 = optional enhancement subject to separate client approval. Details such as calculation formulas, limits and country-specific rules must be signed off with worked examples.

5. Business Rules, Validation & Data Integrity

- BR-4-01: A workflow instance references an immutable published workflow definition version; new versions do not silently alter active cases.
- BR-4-02: State transitions require explicit authorization and concurrency checks; repeated webhooks/actions are idempotent.
- BR-4-03: Notifications must never disclose sensitive cross-tenant records or content in a shared channel.
- BR-4-04: Files require server-side content type/extension validation, upload size limits and malicious content handling.
- BR-4-05: A document may be downloaded only if current user is authorized for its tenant and linked business object.
- BR-4-06: Webhook payloads must be signed, retryable with bounded backoff, deduplicated and never contain unapproved sensitive fields.
- BR-4-07: Report queries must apply the same tenant and row visibility rules as interactive pages.

6. Roles, Permissions & Segregation of Duties

Client must approve a role-permission matrix at action level (view, create, edit, approve, export, delete). Default-deny access is enforced by Spring Security and service-layer object authorization. UI restrictions never substitute for API enforcement.

Role	Primary scope (proposed)
Workflow Designer	Access to assigned module/actions only; final privileges require client-signed matrix.
Process Owner	Access to assigned module/actions only; final privileges require client-signed matrix.
Approver	Access to assigned module/actions only; final privileges require client-signed matrix.
Department Head	Access to assigned module/actions only; final privileges require client-signed matrix.
Reporting Analyst	Access to assigned module/actions only; final privileges require client-signed matrix.
Integration Administrator	Access to assigned module/actions only; final privileges require client-signed matrix.
Compliance Auditor	Access to assigned module/actions only; final privileges require client-signed matrix.

Control requirements: no unapproved self-approval, separation of preparer/approver where mandated, auditing of privileged changes, and tested cross-tenant isolation for all tenant-owned tables.

7. Business Workflows & User Journeys

- Flow 1: Process owner publishes expense approval version -> submitter creates claim -> approvers decide -> reminder/escalation -> completed audit trail.
- Flow 2: Business event commits -> transactional outbox stores event -> background worker routes email/in-app/webhook -> retries recorded.
- Flow 3: Authorized user uploads contract -> metadata/scan -> owner shares within allowed scope -> authorized download is audited.

- Flow 4: Analyst builds filtered report -> scheduler generates snapshot -> role-authorized recipients access export.

8. UI Screens & User Experience

Screen group	Minimum proposed interface
Configurable Workflow & Approval Engine	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Notifications & Internal Communication	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Documents & Controlled Files	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Advanced Reporting & Integrations	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states

Common UX: accessible input labels, confirmation for irreversible actions, reliable error feedback, mobile-responsive layout, keyboard navigation, loading indicators, localization/currency/timezone where applicable. Wireframes require client approval.

9. Data Model & MySQL Entities

Domain module	Proposed core tables (tenant-scoped unless global)
Configurable Workflow & Approval Engine	workflow_definitions, workflow_versions, workflow_instances, workflow_steps, workflow_actions, workflow_delegations
Notifications & Internal Communication	notifications, notification_preferences, notification_templates, notification_deliveries, outbox_events
Documents & Controlled Files	documents, document_versions, document_links, document_access_events, document_retention_policies
Advanced Reporting & Integrations	report_definitions, report_exports, report_schedules, webhook_endpoints, webhook_deliveries, api_credentials

Schema principles: InnoDB foreign keys and appropriate unique constraints; tenant_id on tenant-owned business rows, UTC event timestamps, decimal types for money, effective-date history where necessary, optimistic versioning/locking for concurrent state changes, indexes based on measured queries, Flyway/Liquibase migrations. Do not silently cascade-delete finance, HR or audit evidence.

Client data decisions: retention, import source format, data stewardship and field-level privacy must be confirmed before database design approval. A detailed ERD, dictionary and migration/backfill plan are deliverables, not presumed existing artifacts.

10. REST API Contract - Illustrative Scope

Method	Proposed path	Access
GET/POST	/api/v1/org/workflows/definitions	Verified tenant role; platform-only where /platform
POST	/api/v1/org/workflows/definitions/{id}/publish	Verified tenant role; platform-only where /platform
GET	/api/v1/org/workflows/instances	Verified tenant role; platform-only

		where /platform
POST	/api/v1/org/workflows/instances/{id}/decisions	Verified tenant role; platform-only where /platform
GET	/api/v1/org/notifications	Verified tenant role; platform-only where /platform
PATCH	/api/v1/org/notifications/{id}/read	Verified tenant role; platform-only where /platform
POST	/api/v1/org/documents	Verified tenant role; platform-only where /platform
GET	/api/v1/org/documents/{id}/download	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/reports	Verified tenant role; platform-only where /platform
POST	/api/v1/org/reports/{id}/exports	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/integrations/webhooks	Verified tenant role; platform-only where /platform

API standards: version /api/v1, DTO validation, consistent error envelope and correlation IDs, OpenAPI documentation, stable pagination/filters, idempotency for critical operations, conflict handling (409), throttling for sensitive endpoints and audit hooks.

11. Non-Functional & Security Requirements

Category	Acceptance baseline / to confirm
Availability	Client to specify hosting/SLA, maintenance window and support hours; no implied uptime warranty.
Performance	Confirm p95 read/write latency and concurrent-user workload using realistic records and tenant distribution.
Security	TLS, authorization on every operation, secure secrets, validated uploads/exports, no password/token logging.
Privacy	Personal and financial data minimization; retention, consent, export and deletion policy signed by client.
Reliability	Backups and restore drill, concurrency testing, idempotent events, transactional state changes.
Accessibility	Keyboard-operable WCAG 2.2 AA target where explicitly agreed; browser matrix for Chrome/Edge/Firefox.
Quality	Unit, integration, API-contract, cross-tenant, regression and E2E tests in CI; security review gates.
Observability	Structured logs, metrics, health checks, alerts and audit trail without sensitive values.

12. In Scope / Explicit Exclusions

Included in Phase 4	Deferred or excluded
Configurable Workflow & Approval Engine: agreed requirements	Visual no-code automation for arbitrary scripts or custom code execution
Notifications & Internal Communication: agreed requirements	Public social chat or video conferencing product
Documents & Controlled Files: agreed requirements	Unrestricted data connectors or marketplace ecosystem
Advanced Reporting & Integrations: agreed requirements	Artificial intelligence/OCR inference or AI document

	classification
Configurable Workflow & Approval Engine: agreed requirements	Any external API integration without vendor-specific requirements

All third-party services, subscriptions, tax/legal assurance, external integrations and migration volumes require written scope and cost approval. AI, ML and generative AI features are explicitly out of scope for EnterpriseOS.

13. Acceptance Criteria & User Acceptance Testing (UAT)

- UAT-P4-01: Published workflow version is immutable and old instances remain on the original version.
- UAT-P4-02: Two approvers submitting the same approval concurrently produce one valid state transition.
- UAT-P4-03: Escalation is applied only after configured SLA and with correct approver access.
- UAT-P4-04: An unauthorized user cannot download another department restricted document.
- UAT-P4-05: Webhook signature and replay protections reject tampered/replayed messages.
- UAT-P4-06: Report totals reconcile with selected transactional source records.
- UAT-P4-07: Suppressed/retried notifications are visible in delivery history.

Client approves UAT evidence with expected result, actual result, test data, screenshot/log reference, tester, date and disposition. Critical/high defects must be fixed or explicitly accepted with documented risk before production release.

14. Development Deliverables & Release Gates

Stage	Deliverables	Exit gate
Discovery	Approved BRD/SRS, policy examples, role-permission matrix, dependencies	Requirements sign-off
Design	Wireframes, architecture notes, ERD/data dictionary, API spec, threat/edge-case review	Design sign-off
Build	React pages, Spring Boot modules, MySQL migrations, tests, audit hooks	Functional demonstration
Verify	Automated CI, security/regression tests, data migration checks, UAT runs	UAT approval
Release	Deployment, rollback/backup procedure, API docs, release notes, handover/training	Go-live authorization

Commercial schedule, milestones and cost remain TBD. They depend on approved scope, policy complexity, third-party licensing, existing-data quality, staff capacity and client review cycles. Changes after sign-off use a documented change request with cost and schedule impact.

15. Assumptions, Risks & Open Decisions

Topic	Risk / resolution
Prior phase dependency	Untested upstream permissions or data structures can invalidate this phase; require regression gate.
Policy ambiguity	Unapproved rule formulas or approval paths cause rework; require signed examples.
Tenant privacy	Cross-tenant and overprivileged exports are release blockers; negative tests required.
Data migration	Legacy fields may be incomplete/duplicated; profile and pilot migration before bulk import.

Performance targets	Capacity and realistic datasets are unknown; benchmark after workload approval.
Compliance & legal	Client owns jurisdictional requirements; engineering does not certify legal compliance.
Scope changes	Use issue/decision log and signed change order before adding new integrations/modules.

16. Client Decisions & Requirements Checklist

☐ Which specific business workflows must be configurable by admins? Answer/reference: _____

☐ What decisions determine routing: amount, branch, category, reporting hierarchy or SLA? Answer/reference: _____

☐ Are approval steps sequential, parallel, delegated or consensus-based? Answer/reference: _____

☐ Which notifications require email vs in-app vs live UI delivery? Answer/reference: _____

☐ What are file type, size, access, retention and virus scanning requirements? Answer/reference: _____

☐ Which exact dashboards, report filters, export formats and refresh cadence are needed? Answer/reference: _____

☐ Approved roles/permission matrix attached

☐ Wireframes, report examples and validation rules approved

☐ Hosting/security/retention/deployment requirements agreed

☐ UAT cases and authorized signatories agreed

17. Approval & Sign-off

Signatures indicate approval of the stated Phase 4 requirements baseline and acceptance conditions only. This document does not establish pricing, implementation warranties, or regulatory certifications.

Role	Name / signature	Date
Client Sponsor	_____	_____
Client Product Owner	_____	_____
Technical Delivery Lead	_____	_____

End of Phase 4 requirements document | Version 1.0 | Status: Client review draft